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Other Cystic Kidney Diseases

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Numerous disorders share kidney cysts as a common feature (Box 47.1).[1] These disorders may be inherited or acquired; their manifestations may be confined to the kidney or expressed systemically. They may present at a wide range of ages, from the perinatal period to old age (Fig. 47.1). The kidney cysts may be single or multiple, and the associated kidney morbidity may range from clinical insignificance to progressive parenchymal destruction with resultant kidney insufficiency.

The clinical context often helps distinguish these kidney cystic disorders from one another. Echogenic, enlarged kidneys in a neonate or infant should raise suspicion about autosomal recessive polycystic kidney disease (ARPKD), autosomal dominant polycystic kidney disease (ADPKD, see Chapter 46), tuberous sclerosis complex (TSC), or one of the many congenital syndromes associated with kidney cystic disease. Reduced kidney function in an adolescent suggests ARPKD or nephronophthisis (NPHP) as possible etiologies. The finding of a solitary cyst in a 5-year-old may indicate a calyceal diverticulum, whereas this finding in a 50-year-old is most compatible with a simple kidney cyst. Kidney stones occur in ADPKD and medullary sponge kidneys (MSKs). For those disorders with systemic manifestations, such as ADPKD, TSC, and von Hippel-Lindau (VHL) disease, the associated extrarenal features may provide other important differential diagnostic clues.

For an increasing number of the single-gene disorders, genetic testing is available in expert laboratories around the world. Genetic testing resources are listed at the National Institutes of Health (NIH) Genetic Testing Registry (http://www.ncbi.nlm.nih.gov/gtr).

AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE

Definition

ARPKD is a severe, typically early-onset form of cystic disease that primarily involves the kidneys and biliary tract. Affected patients have a spectrum of clinical phenotypes that correlate in part with the age at presentation.

Etiology and Pathogenesis

Genetic Basis

All typical forms of ARPKD are caused by mutations in a single gene, *PKHD1,* that encodes multiple alternatively spliced isoforms predicted to form both membrane-bound and secreted proteins. The largest protein product of *PKHD1,* the fibrocystin/polyductin complex (FPC), contains one membrane-spanning domain and an intracellular C-terminal tail. FPC localizes, at least in part, to the primary cilium and the centrosome in kidney epithelial cells. The basic defects observed in ARPKD suggest that FPC mediates the terminal differentiation of the collecting duct and biliary tract. However, the exact function of the numerous isoforms has not been defined, and the widely varying clinical spectrum of ARPKD may depend in part on the nature and number of splice variants that are disrupted by specific *PKHD1* mutations.[2]

Although ARPKD is generally considered to be a genetically homogeneous disease, recent reports describe a few patients with atypical forms of ARPKD that result from mutations in either the *DZIP1L* gene or *CYS1,* the human orthologue of the gene disrupted in the *cpk* mouse, the most widely studied experimental model of ARPKD.[3,4] These children presented with hypertension and large, cystic kidneys; some progressed to end-stage kidney disease (ESKD). Distinguishing features of the disease phenotype in these children include more moderate kidney disease and the absence of clinically apparent liver disease.

Pathogenesis

ARPKD typically begins in utero, and the kidney cystic lesion appears to be superimposed on a normal developmental sequence. The tubular abnormality involves nonobstructive, fusiform, and/or saccular dilation of the kidney collecting ducts. In the liver, defective remodeling of the ductal plate in utero results in persistence of primitive bile duct configurations and the evolution of progressive portal tract fibrosis.[5] The remainder of the liver parenchyma develops normally. The defect in ductal plate remodeling is accompanied by abnormalities in the branching of the portal vein. The resulting histopathologic pattern is referred to as congenital hepatic fibrosis.

ARPKD is one of several kidney cystic diseases that are associated with congenital hepatic fibrosis, prompting these disorders to be described as hepatorenal fibrocystic diseases.[6] The primary cilium appears to play a central role in the pathogenesis of ARPKD and other hepatorenal fibrocystic diseases, so the broader term “ciliopathies” is used for these disorders, in which dysfunction of the cilia-centrosome complex appears to underpin the development of a wide array of phenotypes, including kidney cystic disease.[5]

Epidemiology

The estimated incidence of ARPKD is 1 per 20,000 live births. It occurs more frequently in Whites than in other ethnic populations. A recent U.S.-based study using electronic health record (EHR) data calculated an annualized incidence of 1 in 26,485 live births.[7]

Clinical Manifestations

The clinical spectrum of ARPKD is variable. Most cases are identified either in utero or at birth. The most severely affected fetuses have enlarged echogenic kidneys and oligohydramnios because of poor fetal urine output. These fetuses develop the Potter phenotype, with pulmonary hypoplasia, a characteristic facies, and deformities of the spine and limbs. Pulmonary hypoplasia continues to pose a severe clinical challenge and often dictates postnatal survival.[8] A recent EHR-based analysis estimated perinatal mortality to be about 21%.[7] Kidney function, although frequently compromised, is rarely a cause of neonatal death.

For those who survive the first month of life, the reported mean 5-year patient survival rate is 85% to 90%. Morbidity and mortality result from severe systemic hypertension, reduced kidney function, and portal hypertension because of portal tract hyperplasia and fibrosis.[9] Hypertension usually develops in the first few months and ultimately affects 70% to 80% of patients. ARPKD patients have defects in both urinary-diluting capacity and concentrating capacity. Newborns can have hyponatremia, presumably resulting from defects in free water excretion. Although net acid excretion may be reduced, metabolic acidosis is not a significant clinical feature. Retrospective studies have noted an increased incidence of pyuria on urinalysis and culture-confirmed urinary tract infections (UTI).[2]

In the first 6 months of life, ARPKD infants may have a transient improvement in glomerular filtration rate (GFR) because of kidney maturation. Subsequently, a progressive but variable decline in GFR occurs, with patients presenting in the first month of life progressing more rapidly to ESKD than those presenting at more than 1 month of age.[10,11] With advances in effective therapy for ESKD, prolonged survival is common and for many patients, the hepatic complications become the predominant clinical issue.

On average, those infants with serum creatinine values greater than 2.2 mg/dL (200 μmol/L) progress to ESKD within 5 years, but this is highly variable. In longitudinal studies, the probability of freedom from ESKD is approximately 85% at 1 year, approximately 70% at 10 years, approximately 65% at 15 years, and approximately 40% at 20 years.[12]

In children who present later in childhood or in adolescence, portal hypertension is frequently the predominant clinical abnormality, with hepatosplenomegaly and bleeding esophageal or gastric varices, as well as hypersplenism with consequent thrombocytopenia, anemia, and leukopenia. Hepatocellular function is usually preserved. Ascending suppurative cholangitis is a serious complication and can cause fulminant hepatic failure.[13,14]

Other associated features include growth retardation, although the mechanism is not yet defined, and, very rarely, vascular aneurysms (intracranial and extracranial).[15,16]

BOX 47.1 Kidney Cystic Disorders

Disorders

Nongenetic

Developmental

Medullary sponge kidney (MSK)[a]
Kidney cystic dysplasia

- Multicystic dysplasia
- Cystic dysplasia associated with lower urinary tract obstruction
- Diffuse cystic dysplasia: syndromal and nonsyndromal

Acquired

Simple cysts
Solitary multilocular cysts
Hypokalemic cystic disease
Acquired cystic disease (in advanced kidney failure)

Genetic[b]

Autosomal Dominant

Autosomal dominant polycystic kidney disease
Autosomal dominant tubulointerstitial kidney disease
Tuberous sclerosis complex
von Hippel-Lindau syndrome

Autosomal Recessive

Autosomal recessive polycystic kidney disease
Nephronophthisis and nephronophthisis-like disorders (Meckel-Gruber syndrome, Joubert syndrome, Bardet-Biedl syndrome)

X-Linked

Oro-facio-digital syndrome type I

[a]MSK is generally considered to be a sporadic disorder, but recent studies provide evidence for familial clustering, involving autosomal dominant inheritance with reduced disease penetrance and variability in disease expression.
[b]Mendelian Inheritance in Man (http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?db=OMIM)

Pathology

Kidney

The kidney involvement is invariably bilateral and largely symmetric. The histopathology varies depending on the age of presentation and the extent of cystic involvement (Fig. 47.2AB).

In the affected neonate, the kidneys can be 10 times normal size but retain the typical kidney contour. Dilated, fusiform collecting ducts

Age Distribution of Cystic Kidney Diseases

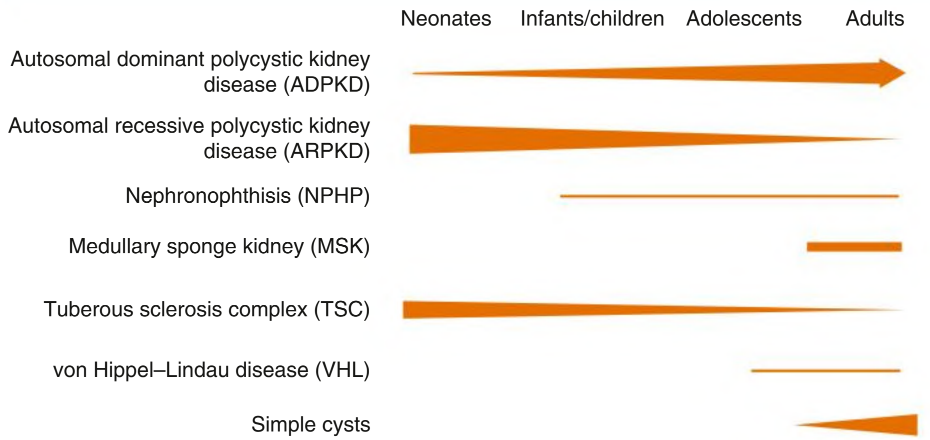

Fig. 47.1 Age distribution of patients with kidney cystic disorders.

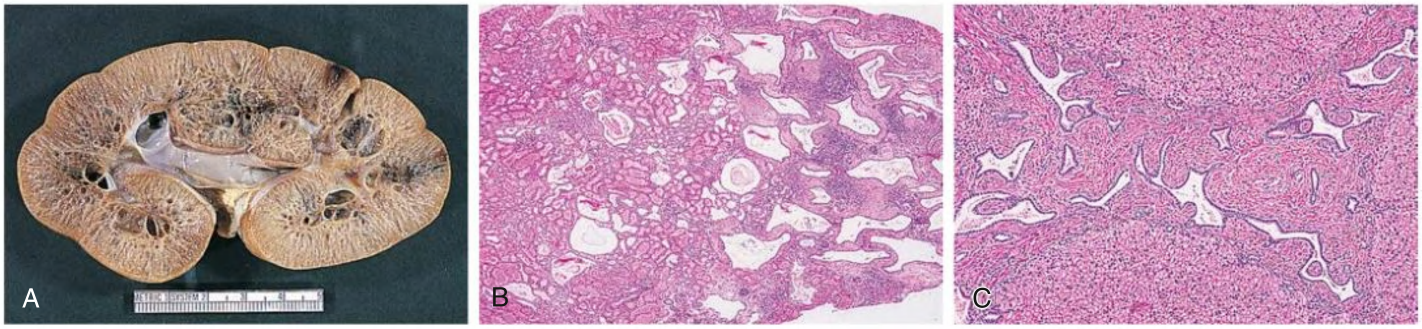

Fig. 47.2 Pathologic Features of Autosomal Recessive Polycystic Kidney Disease. (A) Cut section of autosomal recessive polycystic kidney disease (ARPKD) kidney from a 1-year-old child shows discrete medullary cysts and dilated collecting ducts. (B) Light microscopy shows later-onset ARPKD kidney with prominent medullary ductal ectasia (H&E stain; magnification ×10). (C) Light microscopy of congenital hepatic fibrosis shows extensive fibrosis of the portal area with ectatic, tortuous bile ducts and hypoplasia of the portal vein (H&E stain; ×40).

extend radially through the cortex. On histopathologic sections, the dilated medullary collecting ducts are often cut tangentially or transversely, resulting in a more spheroid appearance. Up to 90% of the collecting ducts are involved. Associated interstitial fibrosis is minimal in neonates and infants but increases with progressive disease.

In patients diagnosed later in childhood, the kidney size and extent of cystic involvement tend to be more limited. Cysts can expand to as large as 2 cm in diameter and may become more spherical. Progressive interstitial fibrosis is probably responsible for secondary tubular obstruction. In older children, medullary ductal ectasia is the predominant finding.

Cysts are lined with a single layer of nondescript cuboidal epithelium. The glomeruli and nephron segments proximal to the collecting ducts are initially structurally normal but are often crowded between ectatic collecting ducts or displaced into subcapsular wedges. The presence of cartilage or other dysplastic elements indicates a diagnosis other than ARPKD, such as cystic dysplasia.

Liver

The liver lesion in ARPKD is characterized by ductal plate malformation. The liver can be either normal in size or somewhat enlarged. Bile ducts are dilated (biliary ectasia) and marked cystic dilation of the entire intrahepatic biliary system (Caroli syndrome) is well described.[13] In neonatal ARPKD, the bile ducts are increased in number, tortuous in configuration, and often located around the periphery of the portal tract. In older children, the biliary ectasia is accompanied by increasing portal tract fibrosis and hypoplasia of the small portal vein branches (see Fig. 47.2C). The hepatic parenchyma may be intersected by delicate fibrous septa that link the portal tracts, but the hepatocytes themselves are seldom affected.

Diagnosis

Imaging

A number of other hepatorenal fibrocystic diseases can resemble or "phenocopy" ARPKD (Table 47.1).[6] Although most of these disorders are characterized by large, echogenic kidneys in the fetus and neonate, these can usually be distinguished by ultrasound. ARPKD kidneys in utero are hyperechogenic and display *decreased* corticomedullary differentiation because of the hyperechogenic medulla (Fig. 47.3A). With high-resolution ultrasound, the radial array of dilated collecting ducts may be imaged. In comparison, ADPKD kidneys in utero tend to be moderately enlarged with a hyperechogenic cortex and relatively hypoechogenic medulla causing *increased* corticomedullary differentiation.

Kidney size typically peaks at 1 to 2 years of age, then gradually declines relative to the child's body size and stabilizes by 4 to 5 years. With age, there is increased medullary echogenicity with scattered small cysts, measuring less than 2 cm in diameter. These cysts and progressive fibrosis can alter the usual kidney contour, causing ARPKD in some older children to be mistaken for ADPKD. Magnetic resonance imaging (MRI) or contrast-enhanced computed tomography (CT) scanning can be useful in delineating the kidney architecture (see Fig. 47.3B). Bilateral pelvicaliectasis and kidney calcifications have been reported in 25% and 50% of ARPKD patients, respectively. In adults with medullary ectasia alone, the cystic lesion may be confused with MSK.

The liver may be either normal in size or enlarged. It is usually less echogenic than the kidneys. Prominent intrahepatic bile duct dilation suggests associated Caroli syndrome. With age, the portal fibrosis tends to progress and in older children, ultrasound typically shows hepatosplenomegaly and a patchy increase in hepatic echogenicity.[13,14]

Genetic Testing

With the identification of *PKHD1* as the principal disease gene, genetic testing is available as a clinical diagnostic tool. The mutation detection rate is 80% to 87%. Current diagnostic approaches include gene-specific analysis and multigene testing panels using next-generation sequencing technologies.[17] Genetic testing is primarily applied in the context of prenatal testing and preimplantation genetic diagnosis.

Patients with two truncating mutations typically have a severe phenotype leading to perinatal or neonatal mortality, whereas missense mutations are typically associated with neonatal survival.[17,18] The European ARegPKD Consortium recently reported that the position of *PKHD1* variants is important in determining the phenotype. Patients with two missense variants affecting FPC amino acids 709 to 1837 or a missense variant in this region combined with a null variant *less frequently* develop ESKD. Patients with missense variants affecting FPC amino acids 1838 to 2624 have less severe liver disease, whereas variants affecting amino acids 2625 to 4074 are associated with poorer hepatic outcome.[19] These findings may facilitate genetic counseling, but caution is warranted because about 20% of ARPKD siblings have discordant clinical phenotypes, perhaps because of the modulating effect of genetic and environmental factors.[17]

Treatment

The survival of neonates with ARPKD has improved significantly in the last two decades because of advances in mechanical ventilation and

TABLE 47.1 **Features of Kidney Cystic Disease in Children**

Clinical Characteristics	ARPKD	NPHP	Meckel-Gruber[a]	GCKD[b]	ADPKD	TSC
Clinical onset (years)	Perinatal	NPHP2/3/13: 0–5 y NPHP: 10–18 y	Perinatal infancy	Infancy and older children	Infancy[c] and older children	Infancy[c] and older children
Enlarged kidneys	Yes	NPHP2: yes NPHP3/13: some cases NPHP: no	Yes	Variable	Occurs	Occurs
Kidney pathology	Multiple cysts	NPHP2/3/13: multiple cysts NPHP: few cysts at C-M junction	Multiple cysts	Multiple cortical cysts	Multiple cysts	Few to multiple cysts angiomyolipoma
Cyst infection	Uncommon	No	Uncommon	No	Occurs	Uncommon
BP	Normal/ increased	NPHP2/13: increased NPHP: normal	Normal	Normal/increased	Normal/increased	Normal/increased
Kidney function (glomerular filtration rate)	Normal/impaired	Normal/impaired	Normal/ impaired	Normal	Normal	Normal
Nephrocalcinosis/ nephrolithiasis	Nephrocalcinosis up to 25%	No	No	No	Nephrolithiases occur	No
Congenital hepatic fibrosis	Yes	Rare	Yes	No	10%–15% infantile ADPKD	No
Pancreas lesions	No	No	No	MODY5	No	No
CNS involvement	No	(Joubert)[d]	Encephalocele; cognitive disability	No	No	Seizures, cognitive disability
Genetics						
Disease gene	*PKHD1*	*NPHP1-NPHP20*	*MKS1-MKS12*	*PKD1* *HNF1B*	*PKD1* *PKD2*	*TSC1* *TSC2*
Genetic testing[e]	Yes	Most	Most	Yes	Yes	Yes

[a]Meckel-Gruber syndrome is a severe, often lethal, autosomal recessive disorder characterized by bilateral kidney cystic dysplasia, biliary ductal dysgenesis, bilateral postaxial polydactyly, and variable CNS malformations. The triad of kidney cystic disease, occipital encephalocele, and polydactyly is most common. Genes disrupted in Meckel-Gruber syndrome have also been identified in NPHP and Joubert patients, suggesting a phenotypic spectrum.
[b]GCKD can occur as the infantile manifestation of ADPKD. Familial hypoplastic GCKD, because of mutations in *HNF1B,* the gene encoding hepatocyte nuclear factor (HNF1β), can be associated with MODY5.
[c]A contiguous germline deletion of both the *PKD1* and *TSC2* genes (the PKDTS contiguous gene syndrome) occurs in a small group of patients with features of TSC, as well as massive kidney cystic disease reminiscent of ADPKD, severe hypertension, and a progressive decline in kidney function with the onset of ESKD in the second or third decade of life.
[d]Joubert syndrome is a genetically heterogeneous, autosomal recessive disorder characterized by developmental defects in the cerebellum (cerebellar vermis aplasia) and the eye (coloboma), as well as retinitis pigmentosa, congenital hypotonia, and either ocular motor apraxia or irregularities in breathing patterns during the neonatal period. The disease can be associated with NPHP, and mutations in several *NPHP* genes have been described in Joubert patients.
[e]Genetic testing listed on the NIH Genetic Testing Registry (http://www.ncbi.nlm.nih.gov/gtr).
ADPKD, Autosomal dominant polycystic kidney disease; *BP,* blood pressure; *C-M,* corticomedullary; *CNS,* central nervous system; *GCKD,* glomerulocystic kidney disease; MODY5, maturity-onset diabetes of the young, type 5; *NIH,* National Institutes of Health; *NPHP,* nephronophthisis; *PKDTS,* polycystic kidney disease with tuberous sclerosis; *TSC,* tuberous sclerosis complex.

other supportive measures. Aggressive interventions such as unilateral or bilateral nephrectomies and continuous hemofiltration have been advocated in neonatal management, but prospective, controlled studies have yet to be performed.[9]

For children who survive the perinatal period, blood pressure (BP) control is a major clinical issue. Angiotensin converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), adrenergic antagonists, and loop diuretics are effective antihypertensive agents. The management of ARPKD children with declining GFR should follow the standard guidelines established for chronic kidney disease (CKD) in children.[9]

Given the relative urinary concentrating defect, children with ARPKD should be monitored for dehydration during intercurrent illnesses associated with fever, tachypnea, nausea, vomiting, or diarrhea. In those infants with severe polyuria, thiazide diuretics may be used to decrease distal nephron solute and water delivery. Acid-base balance should be closely monitored and supplemental bicarbonate therapy initiated as needed.

Close monitoring for portal hypertension is warranted in all patients with ARPKD. There is no correlation between the severity of the kidney and liver disease.[14] Recent studies suggest that the platelet count combined with serial abdominal ultrasound (assessing liver and splenic size) and Doppler flow studies provide good surrogate markers for the portal hypertension severity.[14] Medical management may include sclerotherapy or variceal banding.[13] Surgical approaches such as portocaval or splenorenal shunting may be indicated in some patients. Although hypersplenism is fairly common, splenectomy is seldom warranted. Unexplained fever with or without elevated

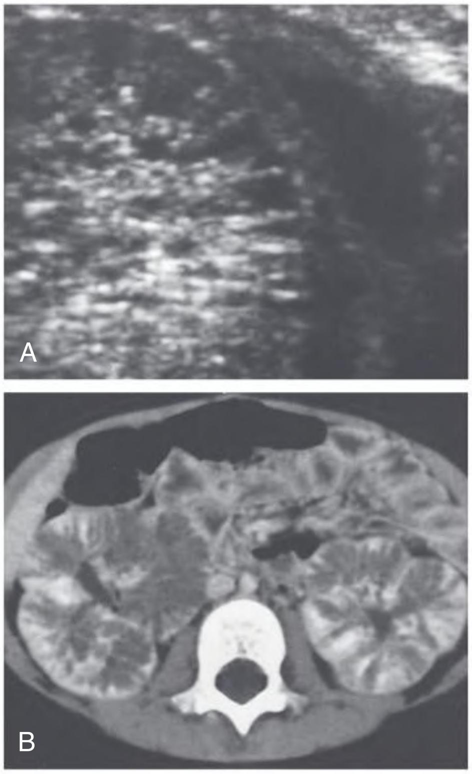

Fig. 47.3 Radiologic Findings Associated With Autosomal Recessive Polycystic Kidney Disease. (A) Autosomal recessive polycystic kidney disease (ARPKD) in a neonate. High-resolution ultrasound reveals radially arrayed dilated collecting ducts. (B) ARPKD in a symptomatic 4-year-old girl. Contrast-enhanced computed tomography shows striated nephrogram and prolonged corticomedullary differentiation.

transaminase levels suggests bacterial cholangitis and requires meticulous evaluation to make the diagnosis and guide aggressive antibiotic therapy.

Effective management of systemic and portal hypertension, coupled with successful kidney replacement therapy (KRT), has allowed long-term patient survival. Therefore, the prognosis in ARPKD, particularly for those children who survive the first month of life, is far less bleak than popularly thought and aggressive medical therapy is warranted.

Although current treatment of ARPKD is entirely supportive, preclinical studies suggest future benefit from new, targeted therapies.[2]

Transplantation

A prolonged period of dialysis in childhood has been associated with both cognitive and educational impairment.[20] Therefore, kidney transplantation is the treatment of choice for ESKD in ARPKD patients. ARPKD is a recessive disorder, and therefore either parent may be a suitable kidney donor. However, the identification of subtle kidney and liver abnormalities on ultrasound in ARPKD parents[21] mandates more extensive posttransplant follow-up. Native nephrectomies may be warranted in patients with massively enlarged kidneys to allow allograft placement.

In some patients, combined kidney-liver transplantation may be appropriate.[22] Indications include the combination of kidney failure and either recurrent cholangitis or significant complications of portal hypertension (e.g., recurrent variceal bleeding, refractory ascites, and the hepatopulmonary syndrome). In addition, liver transplantation may be a consideration for patients with a single episode of cholangitis in the context of Caroli syndrome.[13]

NEPHRONOPHTHISIS: AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE

Definitions

NPHP and autosomal dominant tubulointerstitial kidney disease (ADTKD; formerly known as medullary cystic kidney disease [MCKD]) share a triad of histopathologic features: tubular basement membrane irregularities, tubular atrophy with cyst formation, and interstitial cell infiltration with fibrosis. These histopathologically similar disorders differ in their mode of transmission, age of onset, and genetic defects. NPHP is an autosomal recessive disorder that manifests in childhood, whereas ADTKD occurs in adults. NPHP is more common than ADTKD and has been reported as an isolated kidney disease and also in association with systemic manifestations, including retinitis pigmentosa, congenital hepatic fibrosis, oculomotor apraxia, and skeletal anomalies.

Nephronophthisis

NPHP is an autosomal recessive tubulointerstitial nephropathy and one of the most frequent inherited causes of ESKD in children and adolescents.[23] The term *nephronophthisis* derives from the Greek, meaning “progressive loss of nephrons.”

Genetic Basis

Multiple disease-causing genes have been identified. Defects in *NPHP1* account for 20% of NPHP with large, homozygous deletions detected in 80% of affected family members and in 65% of sporadic cases. Mutations in each of the remaining NPHP genes cause 1% to 4% of NPHP-related disease, meaning that almost two-thirds of cases remain genetically unsolved.[24–26]

Clinical disease expression appears to be exacerbated by oligogenic inheritance, in which patients carry two mutations in a single *NPHP* gene as well as a single-copy mutation in an additional *NPHP* gene. In addition, multiple allelism, or distinct mutations in a single gene, appears to explain the continuum of multiorgan phenotypic abnormalities observed in NPHP-related disorders, which include Meckel-Gruber syndrome, Joubert syndrome, and Bardet-Biedl syndrome.[25,27]

Most of the protein products of the NPHP-associated genes are expressed in the cilia-centrosome complex[5]; NPHP is considered a ciliopathy.[25]

Clinical Manifestations

Kidney disease. Three distinct forms of NPHP (infantile, juvenile, and adolescent) were initially described, based on the age of onset of ESKD. In the infantile form, ESKD occurs before 5 years of age, whereas in juvenile NPHP (the most common form), ESKD occurs at a mean age of 13 years. However, recent studies have demonstrated that there is no clear genotype-phenotype correlation for this spectrum of presentations, and these disorders should be referred to with the single designation of NPHP.

Decreased urinary concentrating capacity is invariable in NPHP and usually precedes the decline in GFR, with typical onset between 4 and 6 years of age. Polyuria, polydipsia, and secondary enuresis are common. Salt wasting develops in most patients with kidney impairment, and sodium supplementation is often required until the onset of ESKD. One-third of patients become anemic before GFR is overtly reduced, perhaps because of impaired regulation of erythropoietin production by peritubular fibroblasts. Growth retardation (out of proportion to the degree of GFR reduction) is common.

Slowly progressive decline in GFR is typical of NPHP, with progression to ESKD at a median age of 13 years. Although symptoms can begin after the age of 2 years, they may progress insidiously, so that

15% of affected patients present with ESKD. There is no specific treatment. The disease is not known to recur in kidney allografts. Unlike patients with PKD or MSK, those with NPHP rarely develop flank pain, hematuria, hypertension, UTI, or kidney calculi.

Children with the relatively rare infantile NPHP variant develop symptoms in the first few months of life and rapidly progress to ESKD, usually before the age of 2 years, but invariably by 5 years of age. Severe hypertension is common in this disorder.[28]

Associated extrarenal abnormalities. Extrarenal abnormalities have been described in approximately 15% to 20% of patients with NPHP,[23] the most frequent of which is retinal dystrophy because of tapetoretinal degeneration (Senior-Loken syndrome). Severely affected patients present with coarse nystagmus, early blindness, and a flat electroretinogram (Leber amaurosis), whereas those with moderate retinal dystrophy typically have mild visual impairment and retinitis pigmentosa. Other extrarenal anomalies include oculomotor apraxia (Cogan syndrome), cerebellar vermis aplasia (Joubert syndrome), and cone-shaped epiphyses of the bones. Congenital hepatic fibrosis occurs on occasion in patients with NPHP, but the associated bile duct proliferation is mild and qualitatively different from that observed in ARPKD.

Pathology

In classic NPHP, the kidneys are moderately contracted with parenchymal atrophy causing a loss of corticomedullary demarcation. Histopathologic findings include tubular atrophy with thickened tubular basement membrane, diffuse and severe interstitial fibrosis, and cysts of variable size distributed in an irregular pattern at the corticomedullary junction and in the outer medulla. However, up to 25% of NPHP kidneys have no grossly visible cysts.

In the typical NPHP kidney lesion, clusters of atrophic tubules alternate either with groups of viable tubules showing dilation or marked compensatory hypertrophy. Multilayered thickening of tubular basement membranes is a prominent histopathologic feature (Fig. 47.4). This pattern is not unique, but the abrupt transition from one type of tubular profile to another suggests NPHP. Moderate interstitial fibrosis, usually without a significant inflammatory cell infiltrate, is interspersed among the atrophic tubules. Spherical, thin-walled cysts lined with a simple cuboidal epithelium may be evident at the corticomedullary junction, in the medulla, and even in the papillae. Microdissection studies indicate that these cysts arise from the loop of Henle, distal convoluted tubules, and collecting ducts. Glomeruli may be normal, although some may be completely sclerosed; others show periglomerular fibrosis, and still others have dilation of Bowman's space, suggestive of glomerulocystic kidney disease.[29]

In comparison, infantile NPHP has features of both classic NPHP (such as tubular cell atrophy, interstitial fibrosis, and tubular cysts) and PKD, including enlarged kidneys and widespread cystic involvement.

Diagnosis

In a child with NPHP and reduced GFR, ultrasound typically reveals normal-sized or small kidneys with increased echogenicity and loss of corticomedullary differentiation. On occasion, cysts can be detected at the corticomedullary junction or in the medulla. Thin-section CT scanning may be more sensitive than ultrasound in detecting these cysts.

The pathologic findings in NPHP are not unique; hence in the early stages of the disease, neither imaging nor histopathology can confirm the clinical diagnosis. Molecular testing can be useful; current strategies using multigene panels and next-generation sequencing technologies allow high-throughput mutation detection for known *NPHP* genes, but to date defects in these genes only explain disease in only up to 40% of patients with NPHP-related disorders.[24,25,27]

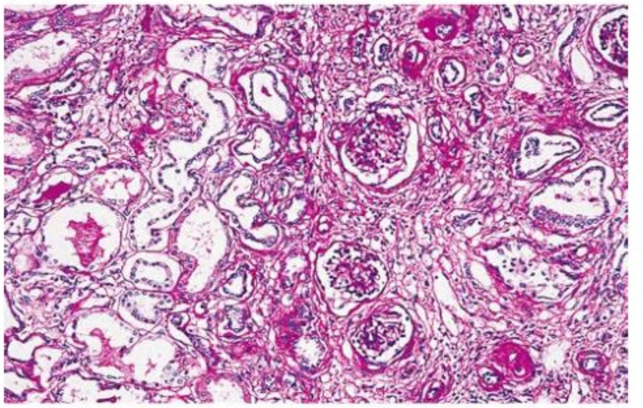

Fig. 47.4 Kidney Pathology in Nephronophthisis. Light microscopy shows tubulointerstitial nephropathy. Atrophic tubules with irregularly thickened basement membranes are surrounded by interstitial fibrosis. Dilated tubules are evident at the corticomedullary junction. (H&E stain; ×40.)

A recent Dutch study of more than 5500 patients with adult-onset ESKD found that 0.5% of the cohort were homozygous for *NPHP1* deletions.[30] A subsequent study in 18 adult patients with kidney biopsies consistent with NPHP found disease-causing mutations in seven patients (39%) involving the *NPHP1, NPHP3, NPHP4,* or *CEP164* genes.[31] These data extend the phenotypic spectrum of NPHP into adulthood and reinforce the utility of genetic testing for patients with ESKD of undetermined etiology.

Treatment

Current treatment of NPHP is supportive but preclinical studies suggest that targeted therapies may be beneficial. For example, treatment with a vasopressin receptor 2 antagonist remarkably attenuated kidney cystic disease progression in a mouse model of *NPHP*-related disease.[32] Others have proposed that therapeutic strategies should target the kidney fibrosis rather than cystic disease in NPHP and note that low-dose paclitaxel has shown promising results in rodent models of kidney fibrosis.[33]

Autosomal Dominant Tubulointerstitial Kidney Disease

The term *autosomal dominant tubulointerstitial kidney diseases* (ADTKD) has recently been proposed for the set of disorders previously described as MCKD, *UMOD*-related kidney disease, familial juvenile hyperuricemic nephropathy (HNFJ1; MIM 162000), and familial glomerulocystic disease with hyperuricemia (MIM 609886).[34] ADTKD is less common than NPHP and the kidney pathologic findings are indistinguishable, although the progressive tubulointerstitial fibrosis resulting in ESKD is slower in ADTKD. Some patients have phenotypically unaffected parents but an affected second- or third-degree relative, suggesting that disease is poorly recognized in affected family members and/or there is variable penetrance.

ADTKD typically manifests in adults, with familial kidney disease characterized by bland urinalysis (i.e., little or no proteinuria or hematuria) and slowly progressive loss of GFR. There may be hyperuricemia or gout. Ultrasound shows normal or small kidneys, sometimes with medullary cysts.

Disease-causing mutations in four genes, *MUC1, UMOD, REN,* and *HNF1B,* are responsible for most cases, with mutations in *SEC61A1* reported in a handful of families.[34] Hyperuricemia is a key feature of disease related to mutations in the *UMOD* and *REN* genes.

Uremia typically occurs after 60 years of age in ADTKD-*MUC1* (MCKD1, MIM 174000). In ADTKD-*UMOD* (MCKD2, MIM 603860) and ADTKD-*REN*, progression to ESKD occurs between the fourth and seventh decades of life. However, disease progression within families can be highly variable, with rare individuals reaching ESKD in their teens, whereas other individuals in the same family may not develop kidney failure until past age 80 years.[35]

MEDULLARY SPONGE KIDNEY

Definition

MSK is characterized by dilated medullary and papillary collecting ducts that give the kidney medulla a spongy appearance. It is associated with nephrocalcinosis, recurrent calcium nephrolithiasis, abnormalities in renal tubular acidification and urinary concentration, bone mineralization defects, and an increased risk for UTI.[36]

Etiology and Pathogenesis

Histopathologic observations of embryonal tissue in the affected papillae, coexistence of kidney tubular defects, and evidence of defects in the genes encoding the RET proto-oncogene and the glial cell line-derived neurotrophic factor (GDNF)[37] suggest that MSK results from a developmental defect in the ureteric bud-metanephric mesenchyme interaction. In addition, MSK occurs more frequently in individuals with other developmental anomalies and/or tumors (e.g., congenital hemihypertrophy, Beckwith-Wiedemann syndrome, congenital anomalies of the kidney and urinary tract [CAKUT] syndrome, and Wilms tumor).[38]

MSK is usually a sporadic disorder, but familial clustering with autosomal dominant inheritance[39] and evidence for genetic alteration in the RET-GDNF axis suggest a genetic basis for MSK in at least a subset of patients.

Epidemiology

In the general population, the frequency of MSK may be underestimated because some affected individuals remain asymptomatic. Up to 20% of patients with nephrolithiasis have at least a mild degree of MSK, but excretory urography in *unselected* patients indicates a disease frequency of approximately 1 in 5000 individuals.

Clinical Manifestations

MSK is asymptomatic unless complicated by nephrolithiasis, hematuria, or infection. Symptoms typically begin between the fourth and fifth decade of life, but adolescent presentations have been reported. Stones or granular debris in patients with MSK are composed of either pure apatite (calcium phosphate) or a mixture of apatite and calcium oxalate. Several factors appear to contribute to stone formation, including urinary stasis within the ectatic ducts, hypercalciuria, and hypocitraturia. Hyperparathyroidism also has been reported.

Hematuria is unrelated to either coexisting stones or infection and may be recurrent. The bleeding is usually asymptomatic unless gross hematuria causes clot-related colic. UTI may occur in association with nephrolithiasis or as an independent event. In patients with stones, infections are more likely to occur in females than in males.

Decreased urinary concentrating ability and impaired urinary acidification are common clinical features. In most patients, the acidification defect is not associated with overt systemic acidosis, but bone mineralization defects are well described.[38]

Pathology

The pathologic changes are confined to the medullary and intrapapillary collecting ducts. Multiple spherical or oval cysts measuring 1 to 8 mm may be detected in one or more papillae. These cysts may be isolated or may communicate with the collecting system. The cysts are frequently bilateral and often contain spherical concretions composed of apatite. The affected pyramids and associated calyces are usually enlarged, and nephromegaly can result when many pyramids are involved. The kidney cortex, medullary rays, calyces, and pelvis appear normal, unless complications (e.g., pyelonephritis or urinary tract obstruction) are superimposed.

Diagnosis

Abdominal plain radiographs often reveal radio-opaque concretions in the medulla (Fig. 47.5B). Historically, the diagnosis was established

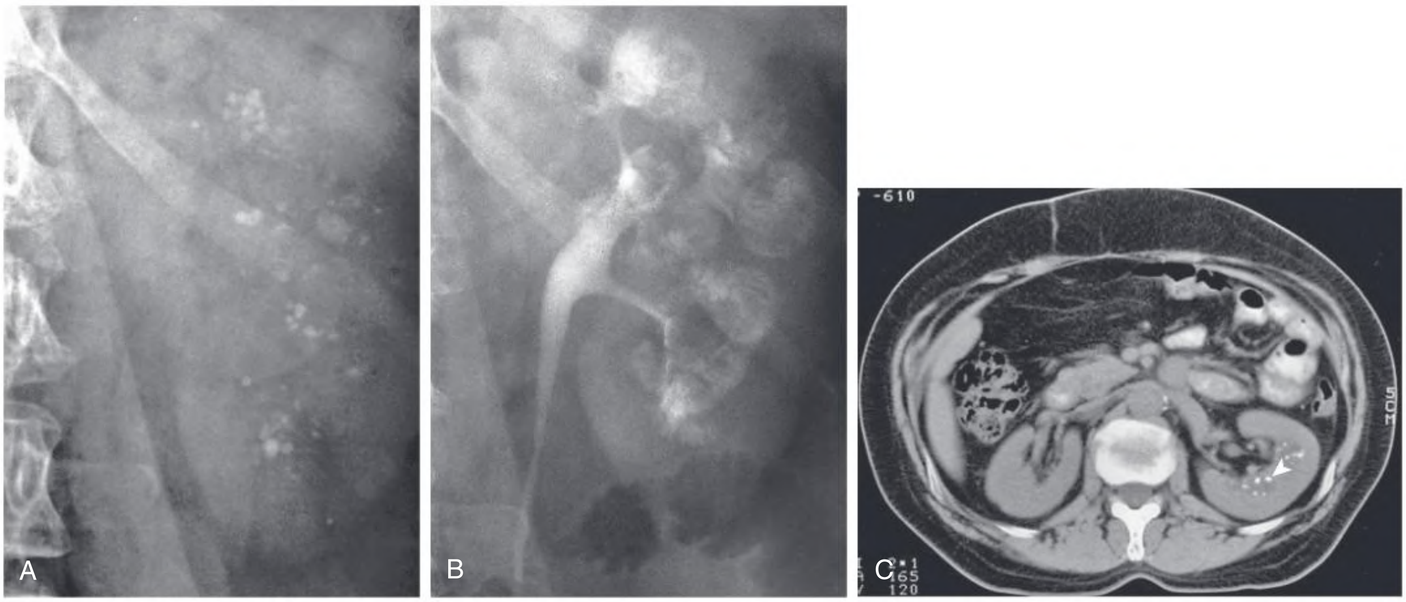

Fig. 47.5 Radiologic Findings Associated With Medullary Sponge Kidney. (A) Preliminary radiograph shows medullary nephrolithiasis. (B) Initial radiograph from excretory urography shows clusters of rounded densities in the papillae among discrete linear opacities (paintbrush appearance). (C) Nonenhanced computed tomography reveals densely echogenic foci in the medulla *(arrowhead)*.

by excretory urography (see Fig. 47.5B). Retention of contrast media by the ectatic collecting ducts appears either as spherical cysts or, more commonly, as diffuse linear striations. The latter imparts a characteristic blush-like pattern to the papillae, the so-called bouquet of flowers or paintbrush appearance. Medullary nephrocalcinosis is a common but not invariant finding. Urography has been almost completely replaced by nonenhanced CT, which may help distinguish MSK from papillary necrosis or even ADPKD (see Fig. 47.5C).

Treatment

Asymptomatic patients with MSK require no therapy. Hematuria in the absence of stones or infection requires no intervention. If the tubular ectasia is unilateral and segmental, partial nephrectomy may alleviate recurrent nephrolithiasis and UTI. However, for the majority of patients who have bilateral disease, medical management is sufficient.

Hypercalciuria and hypocitraturia are the predominant factors contributing to MSK-related nephrolithiasis. The mainstays of treatment are a well-balanced diet consisting of fruits and vegetables, potassium citrate, and high fluid intake.[40] Patients with persistent hypercalciuria and/or recurrent stone formers may benefit from thiazide diuretics. If thiazides are poorly tolerated or contraindicated, inorganic phosphate therapy may be useful. To avoid struvite stone formation, oral phosphates should *not* be used in patients with previous UTIs caused by urease-producing organisms. Patients who recurrently form and pass stones may require lithotripsy or surgical intervention (see Chapter 60).

UTIs should be treated with standard antibiotic regimens and for some patients, prolonged therapy may be warranted. Urease-producing organisms, such as coagulase-negative *Staphylococcus,* are particularly problematic as urinary pathogens in MSK. Positive urine cultures, even with relatively insignificant colony counts, should be vigorously pursued.

With proper management of the clinical complications, the long-term prognosis is excellent. Progression to kidney impairment is unusual.

TUBEROUS SCLEROSIS COMPLEX

Definition

TSC is an autosomal dominant, tumor-suppressor gene syndrome in which benign focal malformations, called hamartomas, develop in multiple organ systems, including the kidneys, brain, heart, lungs, and skin.

Etiology and Pathogenesis

TSC results from inactivating mutations in one of two genes, *TSC1* on chromosome 9q32-q34 and *TSC2* on chromosome 16p13, adjacent to the *PKD1* gene. Large deletions involving both *PKD1* and *TSC2* result in the infantile severe polycystic kidney disease with tuberous sclerosis (PKDTS) contiguous gene deletion syndrome.

The focal development of hamartomas and the variability of disease expression even within families suggest that *TSC1* and *TSC2* function as tumor suppressor genes. The tumor suppressor gene paradigm proposes that two successive mutations are necessary to inactivate a target gene and cause tumor formation. The first mutation, inherited and therefore present in all cells, is necessary but not sufficient to produce tumors. A second mutation occurs after fertilization and is required to induce tumor transformation. The inactivating germ-line mutations identified in *TSC1* and *TSC2,* as well as the loss of heterozygosity detected in *TSC2*-associated and *TSC1*-associated hamartomas, support the hypothesis that both *TSC1* and *TSC2* function as tumor suppressor genes.

The *TSC2* gene product, tuberin, interacts with hamartin, the product of the *TSC1* gene. The hamartin/tuberin (*TSC1*/*TSC2*) complex functions in multiple cellular pathways, primarily by inhibiting the kinase activity of the mammalian target of rapamycin (mTOR), which functions in a protein complex (mTORC1) to integrate nutrient uptake, cell-cycle progression, cell growth, and protein translation[41] (Fig. 47.6). The *PKD1* gene product, polycystin-1, plays a key role in regulating mTORC1 activity by complexing with tuberin and mTOR, thereby inhibiting the mTOR pathway.[42] In a normal adult kidney, mTOR is inactive. With loss of function of either polycystin-1 or tuberin, mTOR activity is upregulated, contributing to dysregulated cell growth and cystogenesis. In addition, hamartin appears to function via TORC1-independent pathways to regulate the structural integrity of the primary cilium, suggesting that ciliary dysfunction is an additional mechanism in TSC pathogenesis.[43]

Epidemiology

TSC affects 1 in 6800 to 15,000 individuals,[44] with approximately 2 million people affected with TSC worldwide.[41] The disease penetrance is quite variable. About 70% of TSC patients are sporadic cases with no family history and the disease apparently results from new mutational events. Mutations in *TSC2* are detectable in approximately 70% of all patients with TSC, whereas *TSC1* mutations can be detected in about 20% of patients. In a substantial fraction of the remaining 10% of patients, the genetic defect may involve genetic mosaicism, whereby a mutational event in either *TSC2* or *TSC1* occurs after the one-cell stage of embryonic development.[41]

Clinicopathologic Manifestations

The clinical features of *TSC1*- and *TSC2*-linked disease are similar, although *TSC2*-linked disease tends to be more severe. The most common clinical manifestations include seizures, cognitive disability and/or autism, skin lesions, interstitial lung disease, and tumors in the brain, retina, kidney, and heart. In affected individuals over 5 years of age, the most common skin lesions are facial angiofibromas (Fig. 47.7), hypomelanotic macules, and ungual fibromas.

Kidney involvement occurs frequently in TSC, with kidney lesions that include angiomyolipomas (AMLs; 85%), cysts (30%–47%), and renal cell carcinoma (RCC; 2%–3%).[45–47] Other kidney neoplasms, interstitial fibrosis with focal segmental glomerulosclerosis (FSGS), glomerular microhamartomas, and peripelvic and perikidney lymphangiomatous cysts also have been reported. Kidney involvement in TSC often progresses insidiously but can result in considerable morbidity, including retroperitoneal hemorrhage, reduced GFR (~1%), and death. Kidney complications are the most frequent cause of death in TSC.[41]

Kidney Angiomyolipomas

Angiomyolipomas (AMLs) are benign kidney hamartomas composed of abnormal, thick-walled vessels and varying amounts of smooth muscle-like cells and adipose tissue (Fig. 47.8AB). These are the most common kidney lesion in TSC patients, detected in approximately 20% of children with TSC before 2 years of age and evident in approximately 67% of patients by age 10 years.[41] Whereas solitary AMLs are found in the general population, TSC-associated AMLs are multiple and bilateral, rarely occurring before 5 years of age, but increasing in frequency and size with age. They can become locally invasive, extending into the perirenal fat or, more rarely, the collecting system, renal vein, and even the inferior vena cava and right atrium.

Clinical manifestations occur secondary to hemorrhage (intratumoral or retroperitoneal) or mass effects (abdominal or flank masses and tenderness, hypertension, reduced GFR). Females tend to have

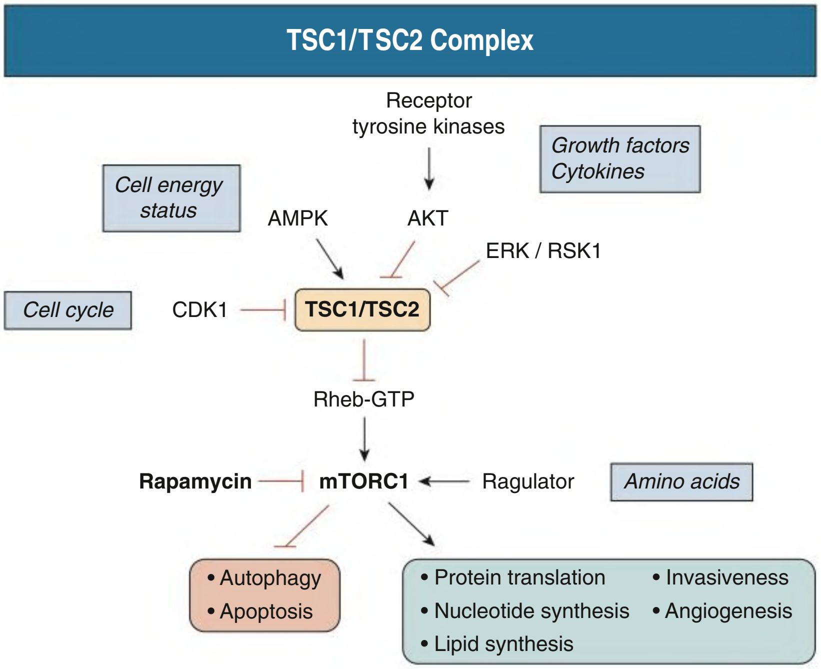

Fig. 47.6 The *TSC1/TSC2* complex regulates mTORC1 signaling by integrating extracellular and intracellular signals that promote metabolic homeostasis. Hamartin *(TSC1)* and tuberin *(TSC2)* integrate signals from extracellular cytokines and growth factors (via AKT and ERK/RSK1), the intracellular energy status (via AMPK), and the cell cycle (via CDK1) to direct signaling pathways that regulate cellular proliferation, differentiation, and migration. Tuberin contains a guanosine triphosphatase (GTPase)-activating protein domain in its carboxy terminus, and when it forms a complex with hamartin, the small GTPase Rheb is converted from its active GTP-bound state to an inactive guanosine diphosphate–bound state. Rheb is an activator of the mTORC1 kinase, which regulates a number of processes linked to protein synthesis and cell growth. mTORC1 is activated physiologically in response to growth factor signaling, which causes phosphorylation of tuberin, dissociation of the hamartin/tuberin complex, and increased levels of Rheb-GTP. Inactivation of the hamartin/tuberin complex through mutations in *TSC1* or *TSC2* leads to inappropriate activation of mTORC1. *AMPK,* Adenosine monophosphate–activated protein kinase; *CDK1,* cyclin-dependent kinase 1; *ERK,* extracellular signal–regulated kinase; *RSK,* ribosomal protein S6 kinase α3. (Modified from Lam HC, Siroky BJ, Henske EP. Renal disease in tuberous sclerosis complex: Pathogenesis and therapy. *Nat Rev Nephrol.* 2018;14[11]:704–716.)

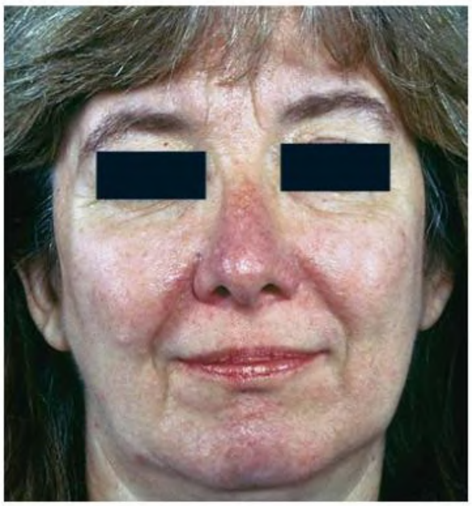

Fig. 47.7 Facial Angiofibromas in Tuberous Sclerosis Complex. The angiofibromas are small red bumps that give a facial rash in a butterfly distribution and on the chin.

more numerous and larger AML than males. Pregnancy appears to increase the risk for rupture and hemorrhage.[47]

Kidney Cystic Disease

Kidney cysts occur less frequently than AML and are evident in 35% to 50% of patients with TSC. However, like AML, kidney cysts tend to increase in size and number over time. The concurrence of cysts and AML is easily detected by CT and strongly suggests TSC.

The cysts in TSC can develop in any nephron segment. When limited in number and size, TSC-related cysts are predominantly cortical. In some cases, glomerular cysts predominate.[29] The cystic lining epithelia appear to be unique to TSC, with large and acidophilic epithelia containing large hyperchromatic nuclei with occasional mitotic figures (see Fig. 47.8C). Associated papillary hyperplasia and adenomas are common.

A small subset of affected infants can present with massive kidney cystic disease, severe hypertension, and a progressive decline in kidney function with the onset of ESKD in the second or third decade of life. The majority of these patients have a contiguous germ-line deletion involving both the *PKD1* and *TSC2* genes, the PKDTS contiguous gene

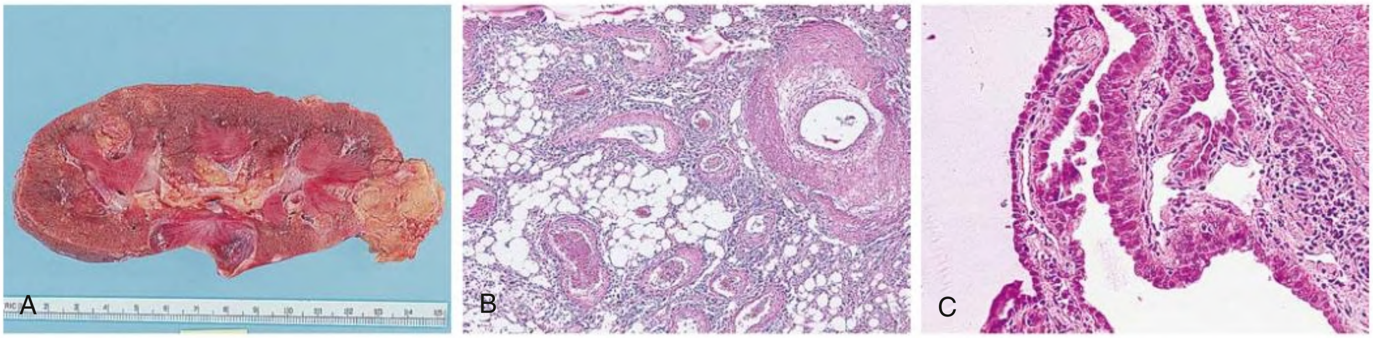

Fig. 47.8 **Kidney Pathology in Tuberous Sclerosis Complex (TSC).** (A) Cut section shows multiple kidney angiomyolipomas. (B) Light microscopy shows angiomyolipoma containing adipose tissue and spindle smooth muscle–like cells interspersed between abnormal vessels with thickened walls (H&E stain; ×16). (C) Light microscopy shows TSC cysts lined with distinctive epithelia consisting of large, acidophilic cells with hyperchromatic nuclei (H&E stain; ×65).

syndrome.[48] Early detection, strict BP control, initiation of mTOR1 inhibitor therapy to limit the growth of AML, and surveillance for TSC-associated renal cell carcinoma (RCC) may improve the overall prognosis.[49]

Kidney Neoplasms

Benign epithelial tumors, such as papillary adenomas and oncocytomas, are common in TSC patients and can be multifocal. Malignant transformation is rare, with 2% to 4% lifetime risk of developing RCC.[41] TSC-associated RCCs have a unique constellation of clinic-pathologic features, including female predominance, younger age at diagnosis, association with AML, multiplicity, and slow progression.[50] Three distinct histopathologic patterns predominate, with papillary RCC being most common, followed by hybrid oncocytic/chromophobe tumor and RCC with granular eosinophilic morphology. Although TSC-associated RCCs tend to have an indolent clinical course, the prognosis compared with sporadic RCC in the general population is unknown.

Diagnosis

TSC is a pleiotropic disease in which the size, number, and location of the lesions can be variable, even among affected individuals within the same family. Major and minor criteria have been developed to guide clinical diagnosis (Box 47.2). Two major features, or one major and two minor ones, are required to make the diagnosis of TSC. Imaging is the mainstay for diagnosis of TSC-associated kidney lesions. The presence of small cysts and fat-containing AMLs is strongly suggestive of TSC.

Kidney imaging is advised for monitoring disease progression in patients with TSC.[47] Ultrasound has historically been the preferred modality; it has high sensitivity in detecting the fat-rich AMLs and kidney cysts but is relatively insensitive for detecting fat-poor lesions. Current recommendations are to perform abdominal CT (Fig. 47.9) or MRI every 1 to 3 years for life to assess AML progression and kidney cystic disease.[51] On occasion, the distinction between an AML and RCC cannot be reliably established by imaging and biopsy is indicated.

TSC-associated kidney cysts can radiologically mimic simple cysts and, when numerous, ADPKD. In the absence of AML, TSC-related kidney cystic disease is suggested by the limited number of cysts compared with ADPKD and the absence of associated hepatic cysts. Although 10% of patients with TSC have hepatic AML, hepatic cysts are rare.

Gene-based diagnosis is currently available for *TSC1*- and *TSC2*-related disease and to detect large-scale deletions associated with the PKDTS contiguous gene syndrome.

BOX 47.2 Clinical Diagnostic Criteria for Tuberous Sclerosis Complex[a]

Major

Facial angiofibromas or forehead plaque
Nontraumatic ungual or periungual fibroma
Hypomelanotic macules (>3)
Shagreen patch (connective tissue nevus)
Multiple retinal nodular hamartomas
Cortical tuber(s)
Subependymal nodule
Subependymal giant cell astrocytoma
Cardiac rhabdomyoma(s) (≥1)
Lymphangioleiomyomatosis
Kidney angiomyolipoma

Minor

Multiple, random dental pits
Hamartomatous gastrointestinal or rectal polyps
Bone cysts
White matter radial migration lines
Gingival fibromas
Non-kidney hamartoma
Retinal achromic patch
“Confetti” skin lesions
Multiple kidney cysts

[a]Two major features or one major feature with two minor features indicate definite TSC; one major feature and one minor feature indicate probable TSC; and one major feature or two minor features indicates possible TSC.

From Roach ES, Gomez MR, Northrup H. Tuberous sclerosis complex consensus conference: revised clinical diagnostic criteria. *J Child Neurol.* 1998;13[12]:624–628.

Treatment

Kidney Angiomyolipomas

Kidney AMLs are generally benign and require no treatment. Larger AMLs frequently develop microaneurysms and macroaneurysms and are associated with a risk of serious hemorrhage, particularly in actively growing AML more than 3 cm in diameter or those less than 5 mm in diameter with aneurysms.[41,47] Large AMLs require preemptive treatment either with nephron-sparing surgery or embolization.[52] In addition to size and complications such as pain or hemorrhage, the inability to exclude an associated RCC should prompt intervention.

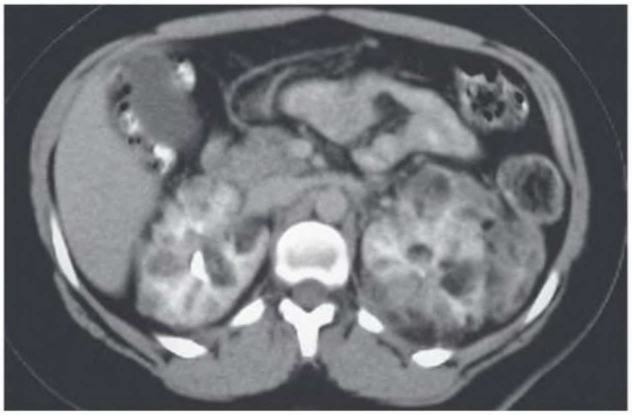

Fig. 47.9 Contrast-enhanced computed tomography showing bilateral angiomyolipomas associated with tuberous sclerosis complex.

When an associated malignancy cannot be excluded, nephron-sparing surgery, such as enucleation or partial nephrectomy, is preferred.

Defective mTORC1 signaling is a central feature of TSC. Recent international TSC guidelines recommend mammalian target of rapamycin (mTOR1) inhibitors (rapalogs) as the first-line therapy for the treatment of asymptomatic, growing angiomyolipomas more than 3 cm in diameter.[47] Sirolimus and everolimus are effective in reducing AML volume and/or postembolization response rates, with an acceptable safety profile,[41] although they have a range of adverse events which must be borne in mind (see Chapter 106).[53]

The increased frequency and size of the AMLs in females and the reports of hemorrhagic complications during pregnancy suggest that female sex hormones may accelerate AML growth. Therefore, it is prudent to caution females with multiple AMLs about the potential risks of pregnancy and estrogen administration.

Kidney Cystic Disease

The mainstay of treatment of TSC-associated kidney cystic disease is strict BP control. Benefit from mTORC1 inhibitors has been suggested by small studies but not definitively demonstrated.[41]

Kidney Carcinoma

TSC-associated RCC should be suspected in enlarging, fat-poor lesions or when intratumoral calcifications are present and should be confirmed by biopsy. Because TSC-associated RCCs are often slow growing and frequently bilateral, nephron-sparing surgery is the treatment of choice.

Kidney Replacement Therapy

CKD, although rare in tuberous sclerosis, can occur by several different mechanisms, including AML-related parenchymal destruction, progressive kidney cystic disease, interstitial fibrosis, and FSGS. On a large French study, TSC-associated ESKD had a prevalence of 1 per 100 patients with TSC and was more frequent in females (63%), with a mean age diagnosis of 29 years.[54]

A follow-up French national KRT registry study evaluated 99 patients with TSC, who progressed to ESKD primarily because of nephrectomy or repeated AML embolization. The age at initiation of KRT was 48.4 years. Fifty-four patients underwent kidney transplantation after an average of 23 months on dialysis. None of the patients had been treated with an mTORC1 inhibitor before dialysis, leaving open the possibility that reducing the size of AML with mTORC1 inhibitors may lower the number of patients with TSC requiring KRT.[55]

VON HIPPEL-LINDAU DISEASE

Definition

VHL disease is a dominantly transmitted, multisystem cancer predisposition syndrome associated with tumors of the eyes, cerebellum, spinal cord, adrenal glands, pancreas, and epididymis, as well as kidney and pancreatic cysts.[56]

Etiology and Pathogenesis

VHL disease is a tumor suppressor disorder, with disease resulting from a germ-line mutation and a subsequent somatic mutation in the *VHL* gene. In approximately 80% of patients, VHL is familial, whereas disease in about 20% of cases results from de novo mutations. Moreover, *VHL* mutations have been identified in sporadic RCC, implying that *VHL* plays an important role in the pathogenesis of RCC.[56,57]

The VHL protein (pVHL) is a central driver in a finely tuned rheostat system that regulates the response to low oxygen levels.[57] In the normal physiologic state, pVHL functions as part of a multiprotein E3 ubiquitin ligase complex that directs a number of proteins, most notably the alpha subunits of the transcription factor, hypoxia-inducible factor (HIF-α), for destruction via the ubiquination pathway. In cells that lack pVHL, HIF-α subunits are stabilized and bind to HIF-β. The heterodimer then translocates to the nucleus, leading to overexpression of HIF-target genes, which encode proteins that regulate glucose uptake, metabolism, extracellular pH, erythropoiesis, angiogenesis (vascular endothelial growth factor [VEGF] and platelet-derived growth factor B [PDGFB]), and mitogenesis (transforming growth factor β [TGF-β]). This transcriptional dysregulation promotes the pathologic growth and survival of endothelial cells, pericytes and stromal cells and, ultimately, their malignant transformation.[56,57]

Clinical Manifestations

VHL disease has an incidence of 1 in 36,000 live births and has been observed in all ethnic groups. Biallelic *VHL* inactivation leads to increased risk for central nervous system (CNS) and retinal hemangioblastomas, RCC, pheochromocytomas, pancreatic islet cell tumors, endolymphatic sac tumors, and papillary cystadenomas of the broad ligament (females) and epididymis (males). In addition, cystic changes can occur in the kidney and pancreas.[56]

VHL-associated disease clusters in two complexes (Table 47.2). In the initial stratification, patients with VHL disease can be subclassified based on a low risk (type 1) or high risk (type 2) of developing pheochromocytoma. Type 1 disease is characterized by large deletions or truncations of the *VHL* gene, which cause complete inactivation of the gene and result in high levels of HIF activity. Type 2 disease primarily involves missense mutations and is associated with partial activity of pVHL. Type 2 patients can be further subdivided into three subtypes: type 2A, which has a low incidence of kidney lesions; type 2B, which is associated with a high risk for RCC; and type 2C, which is associated only with pheochromocytoma and no other malignancies.[56]

In patients with VHL disease, the lifetime cumulative RCC risk approaches 70%.[58] RCCs are typically multiple and bilateral. Although RCC may manifest with hematuria or back pain, more often detection occurs as an incidental finding on unrelated imaging studies or during a screening protocol. The mean age of symptomatic presentation is 35 to 40 years, although patients have been diagnosed in adolescence. In VHL disease, males and females are equally affected with RCC, in contrast to the male predominance in sporadic RCC. VHL disease-associated RCC metastasizes to the lymph nodes, liver, lungs, and bones and accounts for about 50% of VHL-disease related deaths.

In addition to RCC kidney manifestations of VHL, multiple bilateral kidney cysts are found in 50% to 70% of patients.[56] Kidney cysts arise from tubular cells that have undergone somatic loss of the wild-type allele.

Historically, deterioration in GFR among patients VHL was thought to be unusual. However, a recent international consortium study of 96 patients with VHL and nonmetastatic RCC found that 16% of patients pretreatment and 25% of patients posttreatment (e.g., partial nephrectomy or tissue ablation) had CKD.[59]

Pathology

VHL disease-associated RCCs are mostly of the clear-cell type and usually bilateral and multifocal in distribution. Detailed microscopic examination of VHL disease associated kidney cystic lesions often reveals small foci of carcinoma. VHL disease-associated RCCs tend to have low-grade histology and a better 10-year survival than sporadic RCC. More advanced RCCs can metastasize, and metastatic disease is a major cause of death in patients with VHL.

TABLE 47.2 Classification of von Hippel-Lindau Subtypes

Subtype	Tumor Manifestations
Type 1	hemangioblastoma (CNS, retina), kidney cell carcinoma, pancreatic cysts low risk for pheochromocytoma
Type 2A	hemangioblastoma (CNS, retina), pheochromocytoma, low risk for kidney cell carcinoma
Type 2B	hemangioblastoma (CNS, retina), kidney cell carcinoma, pheochromocytoma, pancreatic cysts
Type 2C	predominantly pheochromocytoma very limited risk of hemangioblastoma and kidney cell carcinoma

CNS, Central nervous system.
From Maher ER, Neumann HP, Richard S. von Hippel–Lindau disease: a clinical and scientific review. *Eur J Hum Genet.* 2011;19(6):617–23; and Wind JJ, Lonser RR. Management of von Hippel-Lindau disease-associated CNS lesions. *Expert Rev Neurother.* 2011;11(10):1433.

Diagnosis

The minimum clinical criteria for the diagnosis of VHL disease in an at-risk individual include the presence of a single retinal or cerebellar hemangioblastoma, or RCC, or pheochromocytoma. Up to 50% of affected individuals with familial VHL disease may manifest only one feature of the syndrome. In presumed sporadic cases (20% of patients), the clinical diagnosis requires two tumors (such as two hemangioblastomas, or a hemangioblastoma and a characteristic visceral tumor).

Molecular analysis of the *VHL* gene is indicated in patients with known or suspected VHL or in at-risk children from VHL families, given that unsuspected, untreated tumors can cause significant morbidity. Presymptomatic genotyping can be useful in determining the phenotypic classification of VHL and used to direct monitoring for a specific subset of tumors. In addition, genotyping can be useful in distinguishing whether a pheochromocytoma has occurred in the context of a VHL type 2 mutation or multiple endocrine neoplasia (MEN) type 2 or is nonsyndromic.[60]

In patients with type 2 VHL disease, annual assessment of BP, measurement of urinary catecholamine metabolites, and abdominal ultrasound imaging should be initiated at the age of 2 years. Abdominal MRI and iodine-131 metaiodobenzyl-guanidine (MIBG) scans are indicated if abnormalities are detected. By age 16 years, all patients with VHL disease should have annual MRI scans of the abdomen. Ultrasound is a useful alternative imaging modality in pregnant persons. Annual assessment should be lifelong. Early detection of kidney disease and a multidisciplinary approach to follow-up can substantially improve survival.

Differential Diagnosis

The differential diagnosis of VHL disease–associated kidney lesions includes several conditions, most notably ADPKD and TSC (Table 47.3). As with VHL, ADPKD affects both sexes with a similar mean

TABLE 47.3 Clinical Characteristics of Kidney Cystic Disease in Adults

	Simple Cysts	ADPKD	MSK	VHL	TSC	Acquired Cystic Disease
Age at clinical onset	>30 y	30–40 y	20–40 y	30–40 y	10–30 y	Chronic kidney disease
Cysts	Single/multiple	Multiple	Multiple	Few, bilateral	Multiple	Multiple
Cyst infection	Uncommon	Common	Common	Uncommon	Uncommon	Uncommon
Tumors	No	Rare	No	RCC, often bilateral	AML/RCC	Common
BP	Normal/increased	Increased	Normal	Normal/increased	Normal/increased	Normal/increased
GFR	Normal	Normal/impaired	Normal	Normal	Normal/impaired	Impaired/ESKD
Nephrolithiasis	No	Common	Common	No	No	No
Liver cysts	No	Common	No	Rare	No	No
Pancreas cysts	No	Few	No	Multiple	No	No
CNS involvement	No	Aneurysms	No	Hemangioblastomas	Seizures, cognitive disability	No
Skin lesions	No	No	No	No	See Fig. 47.9	No
Genetics						
Disease gene	No	*PKD1* *PKD2*	No	*VHL*	*TSC1* *TSC2*	No
Genetic testing[a]	No	Yes	No	Yes	Yes	No

[a]Genetic testing is listed at the NIH Genetic Testing Registry (http://www.ncbi.nlm.nih.gov/gtr).
ADPKD, Autosomal dominant polycystic kidney disease; *AML,* angiomyolipoma; *BP,* blood pressure; *CNS,* central nervous system; *ESKD,* end-stage kidney disease; *GFR,* glomerular filtration rate; *MSK,* medullary sponge kidney; *RCC,* kidney cell carcinoma; *TSC,* tuberous sclerosis complex; *VHL,* von Hippel–Lindau disease.

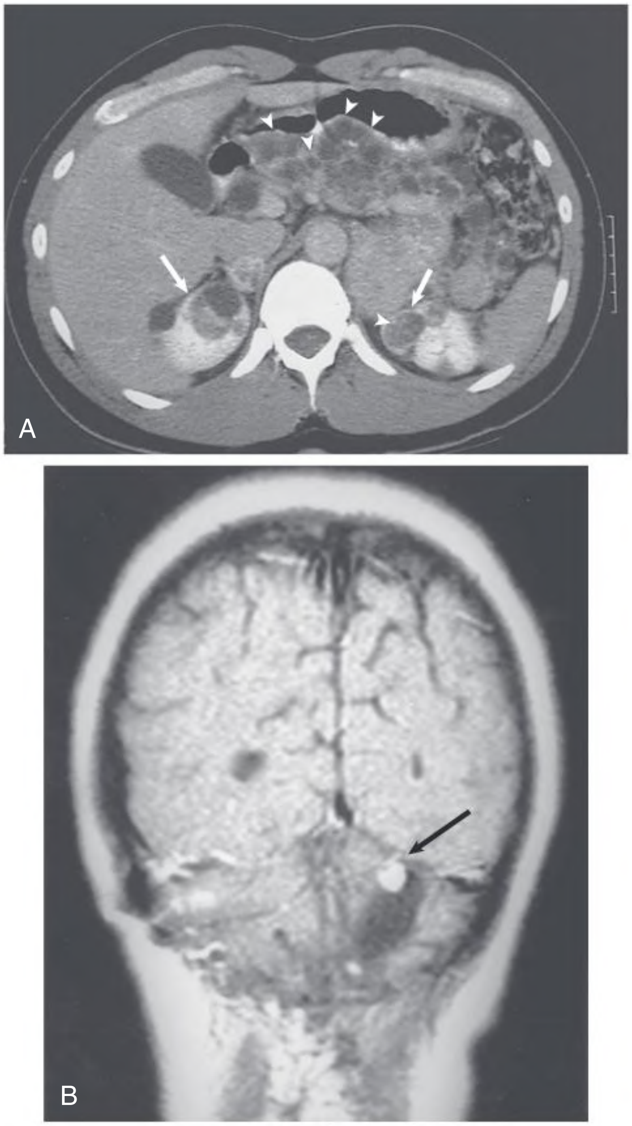

Fig. 47.10 Radiologic Findings Associated With von Hippel–Lindau Syndrome. (A) Noncontrast computed tomography shows massive cystic involvement of the pancreas *(arrowheads)* and bilateral kidney cysts *(arrows)*. (B) Contrast-enhanced magnetic resonance imaging shows a right cerebellar hemangioblastoma with a small enhancing mass *(arrow)*.

age at presentation. However, in VHL disease, kidney involvement is characterized by a few bilateral cysts, RCC, normal kidney size, normal BP, and usually normal GFR. Kidney cyst infection, a frequent finding in ADPKD, is uncommon in VHL disease, and RCC is an infrequent complication of ADPKD. Cysts in the liver are frequent in ADPKD and rare in VHL disease. Pancreatic cysts are rare in ADPKD but can be numerous and scattered through the pancreas in VHL disease (Fig. 47.10A). The CNS in ADPKD is affected by arterial aneurysms, whereas hemangioblastomas are the predominant CNS manifestation in VHL disease (see Fig. 47.10B).

TSC should be considered in the differential diagnosis of multiple kidney tumors. In both TSC and VHL disease, multiple kidney cysts occur. However, the TSC-associated kidney tumor is usually an AML, and extrarenal lesions readily distinguish VHL disease and TSC.

Treatment

Nephron-sparing surgery continues to be the mainstay of treatment for localized RCC in patients with VHL disease.[56] Repeated surgical intervention may be required as tumors continue to develop. Percutaneous and laparoscopic radiofrequency ablation therapy is effective in treating smaller tumors (<3 cm) with low complication rates, but frequent posttreatment monitoring is required.[56] The role of adjuvant therapy with drugs that inhibit the pVHL-HIF-VEGF pathway, such as the multiple tyrosine kinase inhibitors (TKI; e.g., sunitinib, sorafenib, and pazopanib) and monoclonal antibody VEGF inhibitor (e.g., bevacizumab), remains controversial.[58]

Bilateral nephrectomy and kidney transplantation may be an acceptable alternative to repeated nephron-sparing surgery in patients with VHL disease–associated RCC. It remains to be determined whether posttransplant immunosuppression enhances the growth of the retinal and CNS hemangioblastomas and other lesions found in patients with VHL disease.

For RCCs that have locally advanced or metastasized, systemic approaches with immunotherapy (checkpoint inhibitors) and molecularly targeted agents (VEGF receptor TKIs, monoclonal antibody VEGF inhibitors, and mTOR inhibitors), as well as radiation therapy, all may have a role depending on the extent of disease, sites of involvement, and patient-specific factors.[58]

SIMPLE CYSTS

Introduction and Definition

Simple kidney cysts are the most commonly acquired cystic lesion and occur twice as frequently in males as in females. Simple cysts are usually unilateral and may be either solitary or multiple. They occur rarely in children but become increasingly common with age.[61] In a seminal ultrasound study, unilateral cysts were detected in 1.7% of patients aged 30 to 49 years, 11% of patients aged 50 to 70 years, and 22% to 30% of patients older than 70 years.[62] This age-related increase in cyst incidence has been corroborated by MRI studies.

Etiology and Pathogenesis

Simple kidney cysts likely originate from the distal convoluted tubule or collecting ducts and may arise from kidney tubular diverticula, but the pathogenic mechanisms are unknown. Focal tubular obstruction and kidney parenchymal ischemia have both been suggested as etiologic processes. Less likely is the possibility that simple cysts arise from calyceal diverticuli because simple cysts are often found in the kidney cortex and their frequency increases with age.

Age, smoking, reduced GFR, and hypertension[63] have been implicated as risk factors for simple cysts. However, these associations may be coincidental, given that the studies were largely retrospective, the cohorts had variable reasons for diagnostic referral, and the observations were not optimally controlled for patient age.[64]

Clinical Manifestations

Simple cysts are typically asymptomatic. However, increasing evidence supports a relationship between simple kidney cysts and hypertension,[63] which may be associated with increased renalase levels.[65] In addition, red blood cell abnormalities (elevated red blood cell mass, hematocrit, and hemoglobin) are well described in patients with simple kidney cysts.[66] Patients can also present with hematuria, flank pain, evidence of infection, or obstruction of the collecting system. Clinical symptoms are more common with neoplasms than simple cysts and should therefore prompt evaluation for cyst-associated malignancy.[64]

Pathology

Whether unilateral or bilateral, simple cysts are usually spherical and unilocular. They may be solitary or multiple. On average, simple cysts measure 0.5 to 1.0 cm in diameter, but 3 to 4 cm cysts are not uncommon. Simple cysts can occur in the cortex, where they can protrude from the cortical surface (exophytic cysts), the corticomedullary junction, or in the medulla. By definition, they do not communicate with the kidney pelvis. The cyst walls are typically thin and transparent, lined with a single layer of flattened epithelium. Cyst fluid is essentially an ultrafiltrate of plasma. In the wake of infection, cyst walls can become thickened, fibrotic, and even calcified.

Diagnosis

Simple cysts are most often detected as incidental findings during abdominal imaging studies. On occasion, they are discovered during radiologic evaluation of palpable abdominal masses, pyelonephritis, or hematuria after abdominal trauma.

The critical clinical challenge is to distinguish single or multiple simple cysts from cysts associated with ADPKD, other cystic diseases, or RCC. This distinction can usually be made on the basis of patient age, family history, and imaging (see Table 47.3).

The ultrasound features of simple cysts include smooth walls, lack of septae, and no intracystic debris. If the ultrasound pattern is indeterminate, CT scanning should be performed. Benign cysts have homogeneous attenuation, no contrast enhancement, thin and smooth cyst walls, and no associated calcifications, unless prior infection has occurred.

A classification system for kidney cysts based on their appearance and enhancement on CT, described by Israel and Bosniak, is widely used and informs management (see Table 63.5).[67]

Treatment

Simple cysts associated with pain or renin-dependent hypertension can be managed with percutaneous aspiration under radiologic guidance and instillation of a sclerosing agent into the cyst cavity.[68] Laparoscopic or retroperitoneoscopic cyst unroofing (marsupialization) may be more appropriate for large cysts containing volumes in excess of 100 mL. Cyst infection with *Enterobacteriaceae, Staphylococci,* and *Proteus* has been reported and operative or percutaneous drainage is often required in addition to antibiotic treatment.

SOLITARY MULTILOCULAR CYSTS

Solitary multilocular cysts are generally benign neoplasms that arise from the metanephric blastema. These solitary cysts also have been designated multilocular cystic nephroma, benign cystic nephroma, and papillary cystadenoma. By definition, the cystic structures are unilateral, solitary, and multilocular. The cystic locules do not communicate with each other or with the kidney pelvis. These locules are lined with a simple epithelium and the interlocular septa do not contain differentiated kidney epithelia structures.

There is a spectrum of multilocular cysts; at one end is cystic nephroma (CN) and at the other end is cystic partially differentiated nephroblastoma (CPDN), in which the septa contain foci of blastemal cells. It is not clear whether a multilocular cyst represents a congenital abnormality in nephrogenesis, a hamartoma, a partially or completely differentiated Wilms tumor, or a benign variant of Wilms tumor.

There is a bimodal age distribution[69]; approximately half the cases occur in children younger than 4 years and half the cases are detected in adults. The childhood cases (mostly CPDN) are usually found in males, whereas multilocular cysts presenting in adulthood (mostly CN) occur more commonly in females. An abdominal or flank mass is the most common clinical feature because these cysts are typically quite large and often replace an entire pole. Associated hematuria, calculi, urinary tract obstruction, and infection occur in rare instances. Diagnosis can be made either by ultrasound or CT (Fig. 47.11).

Fig. 47.11 Solitary Multilocular Cyst. Contrast-enhanced computed tomography shows a solitary, septated, and well-circumscribed kidney cystic lesion in the right kidney.

Almost all multilocular cysts are Bosniak class III (see Table 63.5), complex kidney cysts suspicious for malignancy.[67] CPDN in children may contain blastema and incompletely differentiated metanephric tissue but usually has a benign course. In adults, associated foci of kidney cell carcinoma or sarcoma must be excluded. For both diagnostic accuracy and treatment, partial nephron sparing surgery is usually required. The typical prognosis of solitary multilocular cysts is excellent.

KIDNEY LYMPHANGIOMATOSIS

Kidney lymphangiomatosis is a rare, generally benign disorder characterized by developmental malformation of kidney lymphatic channels. Synonyms include kidney lymphangioma, peripelvic lymphangiectasia, or kidney hygroma.[70]

The cystic phenotype is widely variable and the underlying pathogenesis is unclear. Dilation may involve a single lymphatic channel or multiple channels. The lymphangiectasis may be unilateral or bilateral, may be limited to the hilar region, or may extend into the kidney parenchyma to the corticomedullary junction. On occasion, kidney lymphangiomatosis may be very extensive and simulate either ADPKD or ARPKD.[71,72] The distinguishing features include cyst lining by lymphatic endothelium and cyst fluid–containing lymphatic constituents such as albumin and lipid.

The characteristic ultrasound or CT findings include multiple, bilateral small peripelvic cysts that splay the renal hilum, as well as capsular cysts in the perirenal space, both separated by thin septations. Kidney lymphangiomas are most often asymptomatic and require no treatment. However, the condition may be exacerbated by pregnancy, resulting in large perinephric lymph collections and ascites that can require percutaneous drainage.

GLOMERULOCYSTIC KIDNEY DISEASE

Glomerular cysts occur in three different clinical contexts: (1) isolated glomerulocystic kidney disease (GCKD); (2) glomerulocystic kidneys associated with heritable malformation syndromes, such as TSC, Meckel syndrome, autosomal dominant tubulointerstitial kidney disease (ADTKD), oro-facio-digital syndrome type I; trisomies 9, 13 and 18; short-rib polydactyly syndromes; and Zellweger cerebrohepatorenal syndrome; and (3) glomerular cystic changes in dysplastic kidneys.[29]

Isolated GCKD can occur as a sporadic condition, a familial disorder, or as the infantile manifestation of ADPKD. Pathologically, the kidney architecture is normal, with no dysplastic elements in the cortex and no evidence of urinary tract obstruction. Cystic dilation predominantly involves Bowman's space and the initial proximal tubule. GCKD is defined as a twofold to threefold dilation of Bowman's space compared with the normal glomerular dimension. Glomerular cysts can be distributed from the subcapsular zone to the inner cortex. The typical ultrasound pattern in GCKD involves increased echogenicity of the kidney cortex with minute cysts, smaller than those evident in ADPKD. T2-weighted MRI typically reveals numerous, uniformly small, cortical cysts.[73] Young infants with either familial or sporadic forms of GCKD may also have kidney medullary dysplasia and biliary dysgenesis.[29]

Heritable GCKD is usually transmitted as an autosomal dominant trait, often occurring as the infantile expression of ADPKD. In these infants, the kidneys are bilaterally enlarged and diffusely cystic. In non–ADPKD-associated GCKD families, the kidneys are typically normal in size, although on occasion, enlarged kidneys are observed. Finally, several sporadic cases of nonsyndromal GCKD have been described, suggesting either new spontaneous mutations or a recessively transmitted disorder.[29]

Familial hypoplastic GCKD (MIM 137920) appears to be a distinct form of GCKD that results from mutations in *HNF1B,* the gene encoding hepatocyte nuclear factor 1β.[74] The kidneys are smaller than normal and often associated with medullocalyceal abnormalities. Tubular abnormalities include hyperuricemia with or without gout, hypokalemia, hypomagnesemia, and polyuria.[75] This multisystem disease is pleiotropic among affected family members with variable associations of hypoplastic GCKD, gynecologic abnormalities, and maturity-onset diabetes of the young, type V (MODY5).

ACQUIRED CYSTIC DISEASE

Hypokalemic Cystic Disease

Kidney cysts are often seen in association with chronic hypokalemia secondary to primary hyperaldosteronism or other kidney potassium-wasting disorders. Nearly 50% of patients with idiopathic adrenal hyperplasia and 60% of patients with adrenal tumors are reported to have kidney medullary cysts, which typically regress after adrenalectomy. Medullary cysts have been reported in children with type I RTA, and patients with mutations involving subunits of the H^+-ATPase can also have radiologic findings of medullary sponge kidney. Whether these abnormalities and/or the severity of nephrocalcinosis alter the progression of CKD in these patients is not known.[76]

Hilar Cysts

Hilar cysts are spherical accumulations of clear, fat droplet-containing fluid within the kidney sinus. These cystic structures are not lined by epithelia. They are most commonly seen in debilitated patients and may represent atrophy of the kidney sinus fat.

Perinephric Pseudocysts

Perinephric pseudocysts are also unlined cavities. They typically occur under the kidney capsule or in the perirenal fascia as a result of urine extravasation from traumatic or spontaneous rupture of a kidney cyst or as the posterior extension of a pancreatic pseudocyst. Surgical intervention is indicated for associated urinary tract obstruction. Otherwise, treatment is directed to the underlying cause.

Acquired Cystic Disease in Kidney Failure

Acquired cystic disease is a significant complication of prolonged kidney failure (see Table 47.3), occurring as the result of progressive structural end-stage kidney remodeling and may be associated with kidney cell carcinoma. Acquired kidney cystic disease is discussed further in Chapter 92.

SELF-ASSESSMENT QUESTIONS

1. Extrarenal complications of ARPKD can include which of the following?
   A Congenital hepatic fibrosis
   B. Caroli syndrome
   C. Growth retardation
   D. Intracranial aneurysms
   E. All of the above
2. Which kidney cystic disease is distinguished by small, contracted kidneys?
   A. ARPKD
   B. NPHP
   C. MSK
   D. von Hippel–Lindau disease
   E. None of the above
3. Which phenotypic feature most accurately distinguishes tuberous sclerosis complex from von Hippel–Lindau disease?
   A. Systemic hypertension
   B. Kidney cysts
   C. Kidney angiomyolipomas
   D. Kidney cell carcinoma
   E. Liver cysts
4. Gene-based testing is a clinically informative tool for which of the following disorders?
   A. ARPKD
   B. NPHP
   C. MSK
   D. von Hippel–Lindau disease
   E. All of the above
5. Medullary sponge kidney is a single-gene disorder in a subset of patients.
   A. True
   B. False

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